

A projection represented by a rank-one operator

MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.exists_nonzero_projection_rankOne_map

theorem

Obtains all projection, normalization, image and compactness conditions on the same witnesses.

Statement

Let A be a nonzero complex C^* -algebra, with no unit assumed. Let H be a separable complex Hilbert space and $\pi : A \rightarrow \mathcal{B}(H)$ a nonzero irreducible representation representing the unique unitary-equivalence class of nonzero irreducible comparisons. There are $p \in A$ and $e \in H$ such that

$$p^* = p = p^2, \quad p \neq 0, \quad \|e\| = 1, \quad \pi(p) = \theta_{e,e},$$

where $\theta_{e,e}(z) = \langle e, z \rangle e$ and the inner product is linear in its second argument. In particular, this same $\pi(p)$ is compact.

Assumptions

Separability concerns H . No prior faithfulness, simplicity, or unit in A is assumed.

Conclusion

The chosen algebra projection has a rank-one orthogonal projection as its represented image, and that image is compact.

Proof route

Obtain a scalar-corner projection, use faithfulness to keep its image nonzero, apply the rank-one lemma, and substitute its image equation in compactness.

▼ Proof steps

1. Apply A nonzero scalar-corner projection to the given singleton π and separable H . It gives $p^* = p = p^2 \neq 0$ with $pap \in \mathbb{C}p$ for all $a \in A$. Apply Faithfulness on the original algebra to obtain injectivity of the same π . Therefore $\pi(p) = 0$ would imply $p = 0$, so $\pi(p) \neq 0$.
2. Apply A scalar corner has a rank-one represented image to this same p , its scalar-corner equations and the given irreducible π . The reason its image has rank one is as follows. The operator $P = \pi(p)$ is a nonzero orthogonal projection. Write $\hat{\pi}(\lambda, a) = \lambda I + \pi(a)$ on $\tilde{A} = \mathbb{C} \oplus A$. If $pap = cp$, then

$$P\hat{\pi}(\lambda, a)P = \lambda P + \pi(pap) = (\lambda + c)P.$$

Choose a nonzero v in the range of P . Then $Pv = v$, so $P\widehat{\pi}(\lambda, a)v = (\lambda + c)v$. The unitized representation is irreducible, hence the orbit of v is dense. Its image under P lies in the closed one-dimensional subspace $\mathbb{C}v$; continuity therefore gives $P(H) \subseteq \mathbb{C}v$. The reverse inclusion follows from $Pv = v$. For $e = v/\|v\|$, the operator P is thus the orthogonal projection onto $\mathbb{C}e$, namely

$$\|e\| = 1, \quad \pi(p)z = \langle e, z \rangle e \quad (z \in H).$$

3. The operator $\theta_{e,e} : z \mapsto \langle e, z \rangle e$ is bounded and has one-dimensional range, so it is compact: the closure of the image of the unit ball is a closed bounded subset of the finite-dimensional space $\mathbb{C}e$. Substituting $\pi(p) = \theta_{e,e}$ gives compactness of the same $\pi(p)$. This does not assert that the entire subspace $\mathbb{C}e$ is a compact set. The same p therefore satisfies all the stated clauses.

The proof obtains a scalar-corner projection, but minimality is not a separate field in the displayed conclusion.

Main citations

- Exact theorem and proof
- A nonzero scalar-corner projection
- Faithfulness on the original algebra
- A scalar corner has a rank-one represented image

Lean source signature (exact)

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theorem exists_nonzero_projection_rankOne_map [Nontrivial A]
  [PartialOrder A] [StarOrderedRing A]
  [TopologicalSpace.SeparableSpace H]
  (pi : NonUnitalCStarRepresentation A H)
  (hsingle : IsSingletonIrreducibleModel.{u, v, u} pi) :
  ∃ p : A, IsStarProjection p ∧ p ≠ 0 ∧
    (∃ e : H, ‖e‖ = 1 ∧ pi p = InnerProductSpace.rankOne ℂ e e) ∧
    IsCompactOperator (pi p)
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In the source	Mathematical meaning
[Nontrivial A]	The algebra is nonzero: $A \neq \{0\}$. This condition does not say whether a unit is assumed; that information comes from the surrounding C^* -algebra structure.
[PartialOrder A] [StarOrderedRing A]	These typeclasses provide the usual order on a C^* -algebra, equivalently $a \leq b$ when $b - a$ is positive. They are used by the state and positivity arguments.
[TopologicalSpace.SeparableSpace H]	The Hilbert space H is separable.
(pi : NonUnitalCStarRepresentation A H)	The representation π of the possibly non-unital algebra.
(hsingle : IsSingletonIrreducibleModel.{u, v, u} pi)	Nonzero irreducibility of π and the universal unitary-comparison condition.

In the source	Mathematical meaning
$\exists p : A, \text{IsStarProjection } p \wedge p \neq 0$	One algebra element is a nonzero self-adjoint idempotent.
$(\exists e : H, \ e\ = 1 \wedge \pi p = \text{InnerProductSpace.rankOne } \mathbb{C} e e)$	For that p , one unit vector e represents its image as the whole operator $z \mapsto \langle e, z \rangle e$.
$\wedge \text{IsCompactOperator } (\pi p)$	The image under that same $\pi(p)$ of the closed unit ball of H has compact closure. This is an explicit additional output clause.

Exact source and proof.

Earlier published Card and PDF

▼ Exact content identity

Declaration: MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.exists_nonzero_projection_rankOne_map

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